

co2 Adoption — Change Report

Branch `co2_env` · commit `5aa45db` · 2 October 2026 · baseline `fa98acb`

This batch took co2 adoption from *tooling wired up* to *tooling has real input*. **No application code changed** — nothing under `artifacts/` or `lib/` was touched. Full findings in `STEP5-FINDINGS.md`; decisions in `CLAUDE.md`.

1 Pre-existing files changed — three, no source

Across both co2 commits, exactly three files that existed before this work were touched. All three changes are additive.

File	Change	What
<code>.gitignore</code>	Ignores	Two blocks: Playwright run output (<code>e2e/.auth/</code> , <code>e2e/.output/</code> , <code>reports</code>) and <code>SECRET.md</code>
<code>package.json</code>	Scripts	Five <code>test:e2e*</code> scripts and the <code>@playwright/test</code> devDependency. One existing line changed — by a trailing comma
<code>pnpm-lock.yaml</code>	Lockfile	The <code>@playwright/test</code> add

No file under `artifacts/` , `lib/` or `scripts/` has ever been modified. Every repo bug found along the way was documented, not fixed — including the 12 now filed as co2 input.

`pnpm-lock.yaml` gets rewritten wholesale if you run pnpm 12.8.1 while the repo is pinned at 12.3.4. That churn is not a real change — `git checkout pnpm-lock.yaml` reverts it.

2 What was added as co2 dependencies

Outside the repo, and easy to miss. These are installed per machine, are not tracked here, and nothing in the repo will tell a new developer they are needed:

- co2-skills plugin, v1.1.12** — 18 skills. The two conductors (`conductor-defect` , `conductor-feature-prepare` / `-develop`) plus `conductor-upgrade-version`; the generators `modelgen-relational` / `-nosql` , `mockgen-tailwind` , four `specgen-*` variants and `testgen-functional`; and the utilities `util-ustagger` , `util-usanalyzer` , `util-projectsync` and `depgen-k8s` .
- ralph-loop plugin** — the persistence layer the conductors start automatically. It re-feeds the same prompt after each session exit, so a long run survives context limits instead of stopping half-finished. Without it installed, a conductor run silently loses that resilience.
- Node 24, pnpm 12.3.4, an own .env , and an own Postgres container.** A shared dev database is not viable — the `scripts/src/clear-*.ts` utilities delete wholesale, so one person's reseed destroys another's conductor run.

Inside the repo:

- `CLAUDE.md` — the project context every skill reads automatically: stack, paths, credentials pointer, and the shell and server rules that make commands work on Windows. Committed, so it loads without being asked for.

- **The context skeleton** — twelve folders, `{qm-pulse,api-server}/context/{model,mockup,specification,test,develop,bug}/`, each held by a `.gitkeep`. These are the output destinations every generator resolves to.
- **Playwright** — `@playwright/test`, `playwright.config.ts` at the root, six specs under `e2e/`, and five scripts. Not optional: both conductors reproduce and verify through it.

Two structural decisions were needed to make co2 fit a monorepo. co2 expects one root folder per application holding both source and context; here source lives at `artifacts/*` with shared `lib/*` packages. The resolution is root-level `<app>/context/` folders holding co2 artifacts *only*, with source untouched — verified to work, since every skill reference resolves under `<app_folder>/context/` and none requires source beside it. Second, `playwright.config.ts` gained a `bugs` project with `testDir: "."` and a layout-agnostic `testMatch`, because `conductor-defect` runs `playwright test <path>` on paths outside the normal `testDir`. That one was a hard blocker, not a tidy-up.

3 PRD.md — the main deliverable

`qm-pulse/context/PRD.md`, 996 lines: 28 modules, 207 user stories, 109 NFRs, 125 constraints, reverse-engineered from the code. Every co2 generator and both conductors read it, so without it Step 6 cannot start.

CHANGE_REQUESTS.md was used for intent only, never as current state. Later CRs revise earlier ones and the register describes things that never shipped — CR014 plans a `pmo` role and two others that do not exist. Summarising it in order would have reintroduced a deleted role into the document that now drives code generation. Every story was confirmed against `artifacts/qm-pulse/src/`.

Tagged `[v1.0.0]` for the existing system and `[v1.1.0]` for one agreed change (§6); otherwise untagged, as `util-ustagger` assigns codes. The usual caveat holds: a reverse-engineered PRD records what the code *does*. Where UI and server disagree it states the *intended* rule, with actual behaviour recorded in the findings file.

4 First real BUG.md — 12 defects

`api-server/context/BUG.md` now holds 12 bugs, all missing or incomplete **server-side authorization**. Each gives an exact request, expected vs actual status, and a `file:line`. Ordered most-severe-first, because `conductor-defect` works in file order. Five were re-read line by line.

Module	N	Most serious
Requirements	2	Any signed-in user can permanently delete any requirement — no project or role check (<code>requirements.ts:737</code>)
Test Execution	2	A normal Save deletes stored rows the client was never sent, destroying rework rows (<code>test-execution.ts:1984</code>)
Test Cases	1	Edit, delete and clone gated on login only (<code>test-cases.ts:580</code>)
Defects	3	Redmine-pulled defects stored with no project, so every user sees them
Calendar	2	Any user can edit or delete any event; <code>createdBy</code> comes from the request body (<code>calendar.ts:105</code>)
Reporting	2	Verdict-report reads <i>and</i> outbound email gated on login only; defect text interpolated into email HTML unescaped

The other ~48 findings — permission mismatches, functional defects, doc drift — were deliberately not filed; they would bury these. `qm-pulse/context/BUG.md` is unchanged and still holds no live bugs.

Do not "tidy" either BUG.md. `conductor-defect` tags and then tries to fix *every untagged* - bullet under a module heading. Worked examples must stay sealed in `<!-- -->`, and preamble text must not use bullets or contain a literal `[vX.Y.Z]` tag.

5 CHANGELOG.md — a deliberate fiction

New, hand-written, marking v1.0.0 as already developed. **Its row names `conductor-feature-develop` although that skill has never run here.** That is intentional: the skill's Redo/Redevelop Guard matches on exactly that string to decide a version is already built. With the row present, a bare `/conductor-feature-develop` `qm-pulse` builds only v1.1.0.

Without it nothing protects the existing app — the older Version Gate skips its check when the version argument is omitted, which is the default. The guard is all that stands between a default invocation and a long attempt to re-implement 207 user stories that already exist. The file explains itself in a note above its separator.

Do not add a `## v1.1.0` section until the 12 defects are fixed. `conductor-defect` rejects a version lower than the highest recorded; the defects are tagged `[v1.0.0]`, so recording v1.1.0 would lock the bug run out. `api-server` needs its own baseline row when it gets a PRD — the guard matches per application.

6 CLAUDE.md — four corrections

Each re-verified in code first. These matter because `CLAUDE.md` loads automatically into every session and is this repo's designated source of truth, so a wrong claim propagates into generated specs.

Claim	Was	Actually
AI provider	OpenAI endpoint	Gemini 2.5 Flash with OpenRouter fallback. No route calls OpenAI — those env vars serve unused audio/image clients
PMO email	HTML + PDF	HTML + inline PNG + optional <code>.xlsx</code> . "pdf" appears zero times in <code>verdict-report.ts</code>
Roles	15, flatly	15 defaults, not an enum — <code>roles</code> is a table with create/rename/delete, so admins can add roles the hardcoded checks will not know
SMTP host	office365	Two conflicting defaults — office365 in <code>lib/email.ts</code> , gmail in <code>verdict-report.ts</code> . Set it explicitly

Version settled at v1.0.0 for the system as deployed; the first new feature is v1.1.0. No CR-to-semver mapping; `package.json` stays `0.0.0`. A decision log now records D1–D9 so settled questions are not re-litigated: CR058–CR077 will *not* be backfilled (17 are cited only in code comments, and the PRD was written from code); CR links stay out of PRD References, so no generator can follow one into stale history; module structure stays at 28.

The one agreed behaviour change (v1.1.0). Configuration → Team Members shows a read-only member list; `qa_lead` views only; the create-user form is limited to tier 3 and above plus `admin`. Today `qa_lead` is

shown that form but the server requires tier 3, so it always fails. Written by hand rather than via `util-updprd`, since that route runs after tagging and tagging is blocked — check for duplicates if it is run later.

7 Open blocker — numeric folder prefix

`util-ustagger` and `modelgen-relational` only resolve root-level folders that *have* a numeric prefix (`1_hub_middleware`). Ours are `qm-pulse` and `api-server`, so both stop with no match — while `util-usalyzer` and both conductors explicitly tolerate no prefix. The restriction is inconsistent across the skill set; confirmed in practice.

So the PRD is **untagged** and `conductor-feature-prepare` cannot complete, since `util-ustagger` is step one of its chain. Either rename to `1_qm-pulse` / `2_api-server` and update the two `Context: paths` in `CLAUDE.md` — safe, as those folders hold only co2 artifacts and packages are named `@workspace/*` — or treat the unprefixed folder as a match per run. **Deferred by decision**, not overlooked.

8 Also in this commit, and removed

- `qm-pulse/context/bug/BUG_MASTER.md` — the zero-bug `conductor-defect` dry run. Kept as the record of which mechanisms were validated, and that `depgen-k8s` was deliberately skipped rather than generating a `Dockerfile` and `k8s/` manifests no fix justified.
- **Removed:** `STEP5-HANDOVER.md` and `STEP5-PROMPT.txt` — scaffolding for the handover that briefed another machine to write the PRD. That transfer happened; anything outliving it was folded into `CLAUDE.md` or this report. Both were untracked, so neither entered history.

Still open: the folder-prefix rename; the 12 defects are filed, not fixed; and two Playwright items — 5 phantom routes in the walk, and a dead `toHaveLength` assertion. Nothing pushed.